Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

Project Title:

" OrbitMind AI– Intelligent Productivity & Study Companion"

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WEEK 02 TASK

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CHAPTER I

Project Proposal

1.1 Project Title

OrbitMind AI — Smart Productivity & Study Assistant

1.2 Problem Statement

Students today juggle classes, assignments, exams, self-study, part-time work, clubs, and personal responsibilities. Even though digital tools exist, most students still struggle with:

- Keeping track of deadlines
- Maintaining a consistent study routine
- Prioritizing tasks properly
- Breaking large subjects into manageable chunks
- Staying focused without distractions
- Getting help quickly without searching everywhere

Traditional productivity apps (like Todolist, Google Tasks, Notion) rely heavily on **manual organization**, which itself becomes a burden. Students often waste time planning rather than studying.

Furthermore, these tools don’t understand the student’s context. They don’t know:

- How heavy a student’s study load is
- What topics are harder
- When exams are near
- The student's productivity rhythm
- How much time a task truly requires

This leads to stress, inconsistency, and burnout.

1.2.1 Why AI is Relevant Here

AI can help solve these problems in ways normal software cannot:

- Analyzes workload patterns
- Understands natural language
- Generates personalized study routes
- Breaks complex subjects into clear steps

- Creates optimized daily/weekly plans
- Provides conversational support through a chatbot

Students no longer need to manually plan everything —

AI acts like a personal academic assistant who guides them intelligently.

OrbitMind AI aims to become a student's daily companion, helping them stay organized, focused, and consistent throughout the semester.

1.3 Objectives

OrbitMind AI is being developed with the following goals:

Primary Objectives

1. Build a centralized smart assistant for student productivity.
2. Support students through AI-generated study plans based on their syllabus and difficulty level.
3. Integrate an AI chatbot for instant academic assistance and task input.
4. Reduce mental load by automating planning and prioritization.

Secondary Objectives

5. Summarize long notes or class content into short, meaningful key points.
6. Provide smart recommendations based on deadlines and workloads.
7. Maintain a clean, distraction-free dashboard that encourages daily usage.
8. Track deadlines, completed tasks, and performance patterns.

The final goal is to help students **study smarter, not harder**, using AI as their companion, not their replacement.

1.4 Proposed Features

Below is a detailed feature list reflecting a realistic, semester-long project scope:

1.4.1 Authentication System

- User Registration & Login
- Password hashing and validation
- Data isolation per user
- Session or JWT-based authentication

A secure authentication system ensures that every student gets a personalized space.

1.4.2 AI Chatbot (OpenAI/Gemini/Grok Powered)

A friendly conversational assistant that can:

- Understand natural language
- Convert user messages into structured tasks
- Recommend study strategies
- Suggest next actions
- Answer academic queries
- Guide the student when overwhelmed

1.4.3 AI Study Plan Generator

Based on the subject/topic, the AI produces a structured study plan:

- Topic breakdown
- Daily or weekly study slots
- Estimated difficulty
- Time allocation
- Revision suggestions

Students can generate study plans before exams, midterms, or regular classes.

1.4.4 Task Management System

- Add/Edit/Delete tasks
- Set deadlines
- Assign priority levels
- Categorize study vs personal tasks
- View daily/weekly/monthly workload

Tasks will sync with AI features to generate recommendations.

1.4.5 AI Productivity Recommendation Engine

This system analyzes:

- Number of tasks
- Types of tasks
- Deadlines

- User’s study habits
- Pending workload

Then it generates recommendations like:

- “Focus on your CSE assignment today.”
- “Break your math chapter into two sessions.”
- “You are overloaded tomorrow; adjust tasks.”

1.4.6 Note Summarizer

Students can paste long notes or class materials.

The AI will generate:

- A short summary
- Key bullet points
- Important highlights

Helps with revision and understanding.

1.4.7 Dashboard

A simple, organized interface showing:

- Tasks
- Deadlines
- AI suggestions
- Recent summaries
- Quick access to AI chatbot

The dashboard becomes the student’s “daily hub”.

1.4.8 Notifications & Reminders

Optional reminders such as:

- Deadline approaching
- Study session suggestion
- Daily plan reminders

1.5. Technology Stack

Frontend

(Selecting simple stack now; may upgrade later)

- HTML
- CSS
- JavaScript

Backend

- Node.js
- Express.js

Database

- Mysql workbench

AI Integration

- OpenAI API / Gemini API / Grok API

Other Tools

- GitHub → Version control
- VS Code → Development
- Postman → API Testing
- Render / Railway / Vercel → Deployment

1.6 Expected Outcome

OrbitMind AI will help students:

- Save time by automating planning
- Reduce stress during exams
- Study smarter with personalized guidance
- Stay consistent with reminders and schedules
- Improve focus through optimized task breakdown
- Understand notes faster using summaries
- Feel supported through a friendly AI chatbot

The final system aims to improve students' academic lifestyle by combining productivity, study planning, and AI assistance into one unified platform. Students no longer need separate apps—**OrbitMind AI becomes their all-in-one academic partner.**

CHAPTER II

2.1 Initial Project Planning

Week 1

- Team formation complete
- Idea finalized: OrbitMind AI
- Researched existing AI productivity tools

Week 2

- Prepared detailed proposal
- Created GitHub repository
- Added team collaborators
- Wrote initial README.md
- Documented planning structure

Week 3–4

- Prepare SRS
- Create use case diagrams
- Design system architecture and workflow diagrams

Week 5

- Build frontend UI prototypes
- Create layout, dashboard mockups

Week 6–7

- Implement backend APIs
- Configure MongoDB Atlas
- Test routes

Week 8

- Integrate AI modules (Chatbot, Summaries, Study Plans)

Week 9–10

- Testing
- Fixing bugs
- Enhancing UI

Week 11

- Deployment on Render/Railway
- Improve performance

Week 12

- Final report
- Presentation slides
- Viva preparation